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CardSat — Calculators, Tools, Games & BASIC

A standalone guide to the interactive and just-for-fun side of CardSat: the three calculators, the ~54-entry Tools menu, the seven games, and Tiny BASIC. It stands apart from the main manual so you can keep it open while you play. Everything here is reachable without a radio, a rotator, or a network connection.

All of this lives under **About** (press **z** from the About screen for Games, or open the **Tools** menu — **t** from Home — for the calculators, BASIC, and the tool set). Throughout CardSat: **;/.** move a selection, **ENTER** confirms, **`** goes back, and **DEL** backspaces in an entry field (it never exits).

Part 1 — The calculators

There are three, all in the Tools menu.

Scientific calculator

A full expression evaluator. Type an expression and press **ENTER**; the result appears and becomes available for the next line. Some things worth knowing:

- **Trig is in degrees**, not radians (so $\sin(30)$ is 0.5). Use **d2r/r2d** if you need to convert.
- **Metric-prefix suffixes** work on numbers — 100k, 2.2n, 47p, 146M. Case matters: M is mega, m is milli. The suffix set runs from f (femto) up.
- Press **** to toggle **engineering-notation output** (a mantissa with a metric prefix, e.g. 4.7 k).
- The **function hints** stay pinned just above the footer, and **'** flips between the general page and a ham-radio page.

Function set. Alongside the usual **sin cos tan atn exp log sqrt abs int** and constants, the calculator carries a batch aimed at radio and orbital work:

- **Math:** **atan2(y,x)** (degrees out), **hypot(a,b)**, **mod(a,b)**, **min/max**, **ncr(n,r)/npr(n,r)**, **sign**, **log2**, **cbirt**, **fact(n)** ($n \leq 170$), **d2r/r2d**, and **rnd()** for a random 0–1.
- **RF:** **swr2rl(s)/rl2swr(dB)**, mismatch loss **mml(swr)**, free-space path loss **fspl(MHz, km)**, noise-figure $\leftrightarrow$ temperature **nf2t(dB)/t2nf(K)**, **dbd(dBi)/dbi(dBd)** (the ± 2.15 dB dipole/isotropic shift), and Doppler **dop(MHz, rrKmS)** — Hz straight from a range rate.
- **Orbital one-liners:** **porb(altKm)** circular period (minutes), **vorb(altKm)** orbital speed (km/s), **fpr(altKm)** footprint radius (km).

Graphing calculator

Plots $y = f(x)$ using the same parser and function set as the scientific calculator, with **x** added as a variable. Press **ENTER** to type a function (**sin(x)**, **x^2-4**, **1/x**, **exp(x/50)**); the curve is sampled one point per pixel column across a pan/zoomable window.

- **Arrow keys pan**, **+/- zoom** about the centre, **a** auto-fits the vertical range to the visible data, **r** resets the window.

- Advanced readouts (see the manual for depth): a **second curve** (Y2), a **trace cursor** with dy/dx , **zero/intersection finding**, **Simpson integration** between marks, a **table view**, and a flat-RAM **CSV plot mode**.
- Trig is in degrees here too, so the default window is sized in degrees.

Programmer calculator

A 64-bit value shown **all at once in hex / decimal / binary / octal**. Enter a number in any base and read it in the others — handy for CI-V addresses, bit masks, and byte math. Bitwise operations and base entry are on-screen; ` returns.

Part 2 — The Tools menu

Open with **t** from Home. Fifty-four entries in six logical bands. **;/.** scroll, a **letter key jumps** to the next tool starting with that letter, **ENTER** opens, ` backs out.

How a form tool works. Most tools are live-recalc forms: a list of input fields with a result area below. **;/.** move between fields, **ENTER** starts editing the selected field (type a number, ENTER commits), and the result **updates as you type** — no “calculate” button. Fields with a **</>** **marker** are *pickers* (cable type, topology, material); use left/right to cycle the choices. Numeric fields accept metric-prefix suffixes (146M, 2.2n). Every form tool **prints** its inputs and results with **p**, and remembers your last-entered values between sessions. ` returns; DEL backspaces in a field.

Below, each tool lists its **input fields** (with default) and what it computes. The four standalone calculators/BASIC are in Parts 1 and 4; the reference/lookup tools (DXCC, CQ/ITU zones, CTCSS, Char lookup, Operating/Radio references) are browsers, not forms — open them and scroll.

Satellite & orbit

- **Location converter** — convert between lat/long, Maidenhead grid, and other position formats. Enter a position, read the equivalents.
- **State vector → GP** — turn an ECI state vector (position + velocity) into general-perturbation orbital elements you can track.
- **CubeSatSim C2C ref** — a reference screen for the AMSAT CubeSatSim “C2C” telemetry format.
- **Conjunction screener** — screens the catalog for close approaches (conjunctions) to a chosen object.
- **Orbital neighborhood** — lists what else is orbiting near a selected satellite.
- **Transponder planner** — plan uplink/downlink around a transponder’s passband.
- **Link margin vs elevation** — shows how received signal margin changes across a pass as elevation (and thus range) varies.
- **Doppler budget (orbit)** — *Apogee alt* (km), *Perigee alt* (km), *Freq* (435.5 MHz). Seeds apogee/perigee from the **active satellite** if one is selected; shows the peak Doppler shift and rate to expect on that frequency.
- **Orbit lifetime (debris)** — *Alt(perigee)* (550 km), *Mass* (4 kg), *Area* (0.03 m²), *Drag Cd* (2.2). Estimates orbital decay lifetime from ballistic coefficient and altitude.
- **Delta-v (Hohmann/plane)** — *Alt 1* (400 km), *Alt 2* (800 km), *Plane chg* (0°). Hohmann-transfer and plane-change Δv between two circular orbits.
- **Pointing loss** — *HPBW* (30°), *Point err* (deg, **seeded from your rotator deadband**). The dB loss from an antenna mispointed by that error at that beamwidth.
- **Polarization / Faraday** — *Freq* (145.9 MHz), *Ionosphere* (picker). Estimates Faraday-rotation of a linear signal through the ionosphere.
- **Debris group screen** — browse a large CelesTrak debris group.

Antennas & feedline

- **Dipole length** — *Freq* (14.2 MHz). Half-wave dipole leg and overall lengths.
- **Vertical / ground plane** — *Freq* (146 MHz). Quarter-wave vertical and radial lengths.
- **Yagi elements** — *Freq* (144.2 MHz), *Elements* (3). Element lengths and spacings for a Yagi of that size.
- **Quad (full-wave loop)** — *Freq* (50.1 MHz), *Elements* (2). Full-wave loop dimensions.
- **Helix antenna** — *Freq* (435 MHz), *Turns* (8), *Circumf* (1.05 wl), *Pitch* (12.5°). Axial-mode helix gain, dimensions, and beamwidth.
- **Coax loss / power** — *Cable* (picker, default LMR-400), *Freq* (146 MHz), *Length* (50 ft), *SWR at load* (1.5). Matched loss, total loss including SWR, and power delivered.
- **Phasing line / stub** — *Cable* (picker), *Freq* (146 MHz), *Length* (picker: ¼/½/... wave fraction). Physical length of a coax section for a wanted electrical length, using the cable's velocity factor.
- **Wavelength / frequency** — *Freq* (146 MHz). Free-space λ and the common quarter, half, and 5/8-wave cut lengths.
- **L/Pi/T match network** — *Topology* (picker), *R source* (50 Ω), *R load* (200 Ω), *Freq* (14.2 MHz), *Loaded Q* (5). Component values for an impedance-match network.
- **Microstrip/stripline Z0** — *Line* (picker), *Er* (4.4), *H sub/gap* (1.6 mm), *W trace* (3 mm), *Freq* (435 MHz). Characteristic impedance of a PCB trace.

RF & measurement

- **Link budget** — a full uplink/downlink signal budget for a satellite path.
- **RF units (dBm/W/V)** — *Power* (100 W). Cross-converts power in W, dBm, dBW, and voltage into 50 Ω .
- **SWR / return loss** — *SWR* (2.0). Return loss, reflection coefficient, and mismatch loss for a given SWR.
- **Free-space path loss** — *Distance* (1000 km), *Freq* (145 MHz). The FSPL in dB.
- **Attenuator pad** — *Atten* (6 dB), *Z0* (50 Ω). Resistor values for pi- and T-topology resistive pads.
- **dB chain sum** — *Stage 1–4* (dB each; +gain, –loss). Sums a gain/loss chain.
- **Cascade NF & G/T** — *Ant gain* (16 dBi), *Sky temp* (150 K), *LNA NF* (0.8 dB), *LNA gain* (20 dB), *Coax loss* (3 dB), *Rig NF* (6 dB). Friis cascade for the antenna→LNA→coax→rig chain: system noise figure, noise temperature, and G/T.
- **Sun-noise G/T measure** — *Y-factor* (1 dB), *Solar flux* (sfu, seeded from the last-fetched 10.7 cm index), *Freq* (435 MHz), *Ant gain* (0 dBi). Turns a measured sun-noise Y-factor into a real G/T figure for your station.
- **IMD products** — *Freq 1* (145.900), *Freq 2* (145.950), *Band low* (145.800), *Band high* (146.000). Lists intermodulation products falling in-band.
- **RF exposure (MPE)** — *Freq* (146 MHz), *Power* (100 W), *Mode duty* (picker), *Ant gain* (2.15 dBi). Estimates the controlled/uncontrolled MPE safe distances.

Electronics & power

- **Complex / polar** — *Real a* (50), *Imag b (j)* (25). Rectangular↔polar conversion for impedance and phasor work.
- **Reactance & resonance** — *Freq* (7 MHz), *Induct L* (10 μ H), *Cap C* (100 pF). Shows X_L , X_C , and the resonant frequency of the L–C pair.
- **RC/RL time constant** — *Resist R* (1000 Ω), *Cap C* (1 μ F). τ and the charge/discharge percentages at 1–5 τ .
- **Battery runtime** — *Capacity* (20 Ah), *RX draw* (0.5 A), *TX draw* (8 A), *TX duty* (30 %), *Usable* (80 %). Operating hours from a battery under a TX/RX duty cycle.
- **Cross-section area** — *Form factor* (picker, default 3U), *Body X/Y/Z* (cm), *Panel area* (m²). Average cross-sectional area for drag/thermal work.
- **Thermal equilibrium** — *Surface* (material picker), *Custom a* (0.25), *Custom e* (0.85). Equilibrium

temperature of a surface in sunlight from absorptivity and emissivity.

- **Trace & wire ampacity** — *Mode* (picker: PCB trace or wire), *Current* (1 A), *Temp rise* (10 °C), *Copper* (1 oz), *Wire* (24 AWG). Safe current / required trace width.
- **Toroid winding** — *Core* (picker, default T50-2), *Target L* (10 µH). Turns needed on that core for the target inductance.
- **PLL / frequency plan** — *Reference* (10 MHz), *R divider* (1), *N divider* (40), *Multiplier* (1). Output frequency and step size of a PLL synthesizer plan.

Terrestrial VHF/UHF/microwave

- **Radio horizon (VHF+)** — *My ant HAAT* (10 m), *Their ant HAAT* (10 m), *k factor* (1.33). Line-of-sight horizon for each station and the maximum LOS path; raise k above ~1.6 to model tropospheric ducting.
- **Fresnel zone clearance** — *Path length* (30 km), *Freq* (144 MHz). First Fresnel zone radius at the path midpoint and the 60% figure that terrain should clear.
- **Tropo ducting index** — *Surface temp* (from weather), *Dewpoint*, *Inversion dT*. A 0–6 ducting-likelihood index from humidity (dewpoint depression) and any temperature inversion. A “watch this” flag, not a forecast.
- **Rain fade (microwave)** — *Freq* (10 GHz), *Rain rate* (25 mm/h), *Path length* (10 km). ITU-style specific attenuation and total path fade for a microwave hop.
- **Terrestrial path budget** — *TX power*, *TX/RX ant gain*, *Line loss*, *Freq*, *Distance*. A two-way link budget: received level and margin over a nominal noise floor, with a workable/marginal verdict. Add tropo/terrain effects separately.
- **Terrain path profile** — *Freq* (for the Fresnel overlay). Set a target grid in **Grid dist/bearing** first, then press f here (WiFi required) to sample ground elevation along the great-circle path; shows the highest obstacle and whether the path clears it with Fresnel margin.

References & lookups (browsers, not forms)

- **DXCC entity lookup** — find a DXCC entity by prefix/name.
- **CQ zones (WAZ) / ITU zones** — the world zone lists.
- **CTCSS tone reference** — the standard sub-audible tone table.
- **Operating references / Radio math reference** — quick operating and formula cards.
- **Char lookup (ASCII/RTTY)** — ASCII, and Baudot/RTTY character codes.
- **Unit converter** — *Value* (100). General unit cross-conversion.

Part 3 — The games

Seven, under **About** → **z (Games)**. Use ;/. to pick and ENTER to start. They’re deliberately small and satellite-themed; **Fn+b** screenshots any of them.

- **Zap the Sats** — a Space-Invaders homage where the invaders are satellites and you defend from below.
- **Doppler Lock** — hold your marker on a frequency that drifts along a Doppler-like S-curve; a tuning-skill trainer.
- **Catch the Pass** — time your move to intercept a satellite as it crosses.
- **Rotor Runner** — a genuine two-axis game: a satellite drifts around the sky and you slew an antenna to keep it centred, exactly like driving a real rotator.
- **Morse Meteors** — letters fall and you clear each by keying its Morse code (two keys for dit/dah); a code-practice game in disguise.
- **Grid Chase** — a Maidenhead grid-square trainer: a location hint appears and you pick the right grid.
- **KESSLER (2-player)** — the headline. A GORILLAS.BAS-style artillery duel on the Moon: two ground stations lob CubeSats over an Earth-with-a-shocked-face, cratering the lunar surface, with wind and selectable gravity (Moon/Mars/Earth).

KESSLER in depth

Local (hot-seat): on the title screen pick the round count (1–9), cycle gravity with **g**, and press **ENTER** to start. Each turn shows a compact aim form on the *aiming player's* side of the screen — **>** marks the active field. Type an **angle**, press **ENTER** to move to **velocity**, type it, press **ENTER** to fire. **DEL** edits digits; **,//** switch fields; **;/** nudge a value. Land a CubeSat on (or next to) the enemy base to score. A shot too slow to leave the pad fizzles and craters in place; arcing one back onto your own roof loses the round.

Over LoRa (two Cardputers): press **n** on the title screen to host — you become Player 1 and beacon an invite on your shared LoRa frequency. Another CardSat running KESSLER joins automatically as Player 2. Both radios build the *same* battlefield from a shared seed and simulate every shot identically, so the game stays in sync while only tiny control packets cross the air. You aim only on your turn; the other station shows “waiting for peer’s shot.” Range is your LoRa link’s range. (This needs two devices, obviously, and uses the same LoRa settings as the messaging screen.)

Part 4 — Tiny BASIC (language reference)

CardSat has a genuine line-numbered BASIC (Tools → Tiny BASIC) that can **read the live satellite/GPS state** and **draw on the screen**. One firm rule: **there is no INPUT and no INKEY\$** — programs don’t take interactive input, by design. System data is read-only and snapshotted when you call for it.

Editing and running

- Type a **numbered line** to store it: `10 PRINT "HELLO"`. Lines run in numeric order.
- Type an **unnumbered statement** to run it immediately.
- Re-enter a line number with new text to replace it; enter a bare number to delete that line.
- Programs are edited on-device; `LPRINT` and the file words below reach storage.

Values and variables

- Numbers are floating-point. Variables are single letters A–Z (and letter+digit where supported).
- **LET v = expr** assigns; the **LET** is optional (`A = 5` works).
- **DIM @(n)** declares the numeric array `@()`, indexed `@(0)...``@(n)`.

Operators

- **Arithmetic:** `+` `-` `*` `/`, unary minus, and `^` for power. Parentheses group.
- **Relational** (in **IF**): `=` `<>` `<` `>` `<=` `>=`. A true comparison is non-zero.
- Standard precedence: power, then `*` `/`, then `+` `-`, then comparisons.

Statements

- **PRINT list** — print numbers/strings/expressions. Separators: `;` joins with no gap, `,` advances to the next column. A trailing `;` suppresses the newline. **PRINT** with nothing prints a blank line.
- **REM text** — a comment to end of line.
- **IF cond THEN stmt** (or **THEN linenum**) — run the statement / jump if the condition is non-zero.
- **GOTO linenum** — jump.
- **GOSUB linenum ... RETURN** — call a subroutine and come back.
- **ON expr GOTO n1,n2,...** — computed jump (1 selects the first target, etc.).
- **FOR v = a TO b [STEP s] ... NEXT v** — counted loop.
- **READ v[,v...] / DATA n1,n2,... / RESTORE** — read constants from **DATA** lines; **RESTORE** rewinds to the first.

- **END** — stop the program.

Numeric functions

ABS(x), SGN(x), INT(x), SQR(x), SIN(x), COS(x), TAN(x), ATN(x), LOG(x), EXP(x), MIN(a, b), MAX(a, b), RND(n) (random). Trig follows the calculator convention.

Output to the printer / files

- **LPRINT list** — send a line to the report sink (same destination the tools print to).
- **File I/O (enable in Settings first):** FOPEN, FPRINT, FCLOSE, and FILES to list. Gated so a program can't touch storage unless you allow it.

Graphics

Drawing goes to the screen and the frame **holds after the program ends**, so you can plot and look. Colour is an optional last argument (a small palette index); coordinates are screen pixels.

- **CLS** — clear the drawing.
- **PSET x, y [, colour]** — set a pixel.
- **LINE x1, y1, x2, y2 [, colour]** — draw a line.
- **CIRCLE x, y, r [, colour]** — draw a circle.
- **TEXT x, y, "string" [, colour]** — draw text.
- **SHOW** — present the drawn frame (flush what you've plotted).

The system bridge (what makes it CardSat's BASIC)

Each of these samples the live state at the moment it's called:

- **SATSEL i** — select catalog satellite *i* and run SGP4 for it (host-verified accurate). The read-outs below then refer to it.
- **TXSEL i** — select a transponder.
- **Selected-satellite position:** SATAZ, SATEL (az/el, degrees), SATLAT, SATLON (sub-point), SATRANGE (km), SATRR (range rate, km/s — feed it to the calculator's dop() for a Doppler shift).
- **Upcoming passes:** PASSAOS(k), PASSLOS(k), PASSMAX(k) — AOS time, LOS time, and max elevation of the *k*-th upcoming pass of the selected satellite.
- **Your station:** GPSLAT, GPSLON.
- **Space weather & propagation:** SFI (10.7 cm flux), KP, AINDEX, SSN (sunspot number), FLARE (0–5 for none/A/B/C/M/X), BZ (solar-wind IMF Bz, nT), SWSPEED / SWDENS (solar-wind speed & density), MUF (estimated MUF, MHz), and FCKP1 / FCKP2 / FCKP3 (SWPC 3-day max-Kp forecast). Read SPWX0K first to know whether space-wx has been fetched (values read 0 when a feed is unavailable).
- **Pointing:** MAGDECL — magnetic declination at your QTH in degrees (east positive), from the built-in IGRF model. Add it to a true bearing to get a magnetic compass heading. Approximate (a few degrees); requires your location to be set.

Because the data is read-only and snapshotted, BASIC can *observe* the sky and plot it, but never drives the radio or rotator.

Two worked examples

Print the next three passes of catalog satellite 0:

```
10 SATSEL 0
20 FOR K = 1 TO 3
30 PRINT "AOS "; PASSAOS(K); "    MAXEL "; PASSMAX(K)
40 NEXT K
```

Plot the selected satellite's position as a dot on a simple az/el field:

```
10 SATSEL 12
20 CLS
30 LINE 0,67,239,67 : REM horizon
40 X = SATAZ * 239 / 360
50 Y = 67 - SATEL * 67 / 90
60 CIRCLE X, Y, 3, 2
70 SHOW
```

Where to go deeper

This guide is the tour. For field-by-field detail on any tool, the full graphing calculator instrument set, BASIC coordinate/colour specifics, and the exact game scoring, see the main **MANUAL.md** (Tools, Games, and BASIC chapters). The LoRa KESSLER wire format is documented in docs/interfaces/LORA_KESSLER_NETPLAY.md.